

Stephen W. McArdle

Software Engineer

📍 Saylorsburg, PA

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Skills

Core Languages

JavaScript, TypeScript, SQL, Python,
Java, C#, Solidity, C/C++

Frontend & Visual Systems

Vue, React, Angular, Vite, Three.js, p5.js,
WebGL, HTML5 Canvas, SVG, CSS3

Backend & Data

Node.js, PostgreSQL, MongoDB, Express,
Spring, Relational Data Modeling, REST
APIs

Infrastructure & Delivery

Docker, CI/CD Pipelines, Git, AWS, Azure,
Linux

Physical Computing & Fabrication

Pen Plotting, Machine Embroidery,
Laser Engraving, Digital-to-Physical
Production Pipelines

Languages

English

Experience

Self Employed

Contract Developer

March 2024 - Present

Remote

- Designed and implemented real-time generative rendering systems using JavaScript, p5.js, and WebGL, emphasizing deterministic execution and high-performance output across thousands of unique variations.
- Built internal tooling and preview workflows to iterate on generative parameters, validate outputs, and ensure visual consistency prior to release.
- Collaborated with traditional artists to translate aesthetic intent into algorithmic systems; currently developing modern frontend applications using React, Vite, and Tailwind CSS.

Generative Artworks

Co-Founder

June 2020 - Present

Remote

- Architected generative art pipelines using JavaScript, p5.js, and Three.js, enabling complex procedural visuals rendered consistently across environments.
- Designed and built Node.js-based tooling for generative content creation and validation; tooling was later acquired and adopted by Art Blocks.
- Released large-scale generative art collections on Ethereum, leveraging both established platforms and custom smart contracts, with \$12M+ in combined primary and secondary market volume.
- Spoke publicly at NFT NYC and Art Blocks Marfa Weekend on generative systems, creative computation, and art on blockchain.

Marsh McLennan

Technical Lead

June 2019 - May 2021

Hoboken, NJ

- Led distributed agile teams building and maintaining data-driven web platforms using MEAN stack, Java/Spring, and PostgreSQL, including relational data modeling for complex business workflows.
- Re-architected legacy applications into modern, modular systems, improving reliability, performance, and developer velocity.
- Built and maintained CI/CD pipelines using Docker and Jenkins, enabling repeatable deployments across environments.

Everest RE

Software Engineer

June 2018 - May 2019

Liberty Corner, NJ

- Designed, developed, and deployed full-stack .NET applications across Azure and on-prem infrastructure.
- Introduced AI-powered automated document processing, significantly reducing manual underwriting workload.
- Collaborated within agile teams to ship reliable, production-grade systems in regulated environments.

Education

Stevens Institute of Technology

Bachelor of Science, Computer Science
3.64 GPA

May 2018

Hoboken, NJ

Minor: Science, Technology, & Society